Sexual Hormones in Humans

Exam code: 0610 & 0970

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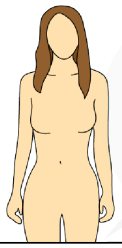
Help me understand

Secondary Sexual Characteristics

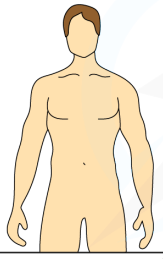
- Primary sexual characteristics are present during development in the uterus and are the differences in reproductive organs etc between males and females
- Secondary sexual characteristics are the **changes that occur during puberty** as children become adolescents
- They are controlled by the release of **hormones** - **oestrogen** in girls and **testosterone** in boys

Human secondary sexual characteristics

Female secondary sexual characteristics:

FEMALE	EFFECTS OF OESTROGEN
	BREASTS DEVELOP
	BODY HAIR GROWS
	MENSTRUAL CYCLE BEGINS
	HIPS GET WIDER

Male secondary sexual characteristics:

MALE	EFFECTS OF TESTOSTERONE
	GROWTH OF PENIS AND TESTES
	GROWTH OF FACIAL AND BODY HAIR
	MUSCLES DEVELOP
	VOICE BREAKS
	TESTES START TO PRODUCE SPERM

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Some changes occur to both boys and girls, including **growth of sexual organs** and **growth of body hair**

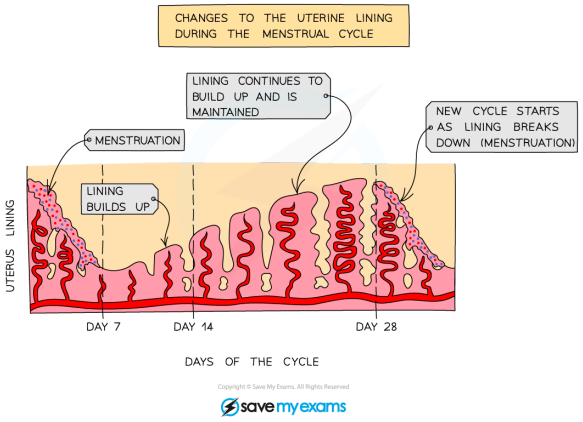
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

swings

Did this video help you? Yes No

The Menstrual Cycle

- Starts in early adolescence in girls (around age **12**) and is controlled by **hormones**
- The average menstrual cycle is **28 days** long
- **Ovulation** (the release of an egg) occurs about **halfway** through the cycle (day **14**) and the egg then travels down the oviduct to the uterus
- Failure to fertilise the egg causes **menstruation** (commonly called a period) to occur - this is caused by the **breakdown of the thickened lining of the uterus**
- Menstruation lasts around **5 – 7 days** and signals the beginning of the next cycle
  After menstruation finishes, the lining of the uterus starts to thicken again in preparation for possible implantation in the next cycle



Changes in the lining of the uterus during the menstrual cycle

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Did this video help you? Yes No

Hormones of the Menstrual Cycle: Extended

- The menstrual cycle is controlled by hormones released from the **ovary** and the **pituitary gland** in the brain

The roles of FSH and LH

Changes in the levels of the pituitary hormones FSH and LH in the blood during the menstrual cycle

- **FSH** (**follicle-stimulating h**ormone) is released by the **pituitary gland** and causes an **egg to start maturing** in the ovary
- It also **stimulates the ovaries** to start releasing **oestrogen**
- The **pituitary gland** is stimulated to release **l**uteinising **h**ormone (**LH**) when **oestrogen** levels have reached their peak
- **LH** causes **ovulation to occur** and also **stimulates the ovary** to produce **progesterone**

Changes in the levels of oestrogen and progesterone in the blood during the menstrual cycle

- **Oestrogen** levels rise from day 1 to peak just before **day 14**
- This causes the **uterine wall to start thickening** and the **egg to mature**
- The peak in oestrogen occurs just before the egg is released
- **Progesterone** stays low from day 1 – 14 and starts to rise once ovulation has occurred
- The increasing levels **cause the uterine lining to thicken further**; a fall in progesterone levels causes the uterine lining to break down (**menstruation / 'period**')

Interaction between all four of the menstrual cycle hormones

- The pituitary gland produces **FSH** which stimulates the development of a **follicle** in the ovary
- An egg develops inside the follicle and the follicle produces the hormone **oestrogen**
- Oestrogen causes **growth and repair of the lining of the uterus wall** and inhibits production of **FSH**
- When oestrogen rises to a high enough level it stimulates the release of **LH** from the pituitary gland which causes **ovulation** (usually around day 14 of the cycle)
- The follicle becomes the **corpus luteum** and starts producing **progesterone**
- Progesterone **maintains the uterus lining** (the thickness of the uterus wall)
- If the ovum is not fertilised, the corpus luteum breaks down and progesterone levels drop
- This causes **menstruation**, where the uterus lining breaks down and is removed through the vagina - commonly known as having a period
- If pregnancy does occur the corpus luteum **continues to produce progesterone**, preventing the uterus lining from

do so throughout the pregnancy

Where hormones involved in the menstrual cycle are made and act

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